

Cyclooxygenase-Dependent Tumor Growth through Evasion of Immunity

SCIFACT-EVIDENCE-7975937

The mechanisms by which melanoma and other cancer cells evade anti-tumor immunity remain incompletely understood. Here, we show that the growth of tumors formed by mutant Braf(V600E) mouse melanoma cells in an immunocompetent host requires their production of prostaglandin E₂, which suppresses immunity and fuels tumor-promoting inflammation. Genetic ablation of cyclooxygenases (COX) or prostaglandin E synthases in Braf(V600E) mouse melanoma cells, as well as in Nras(G12D) melanoma or in breast or colorectal cancer cells, renders them susceptible to immune control and provokes a shift in the tumor inflammatory profile toward classic anti-cancer immune pathways. This mouse COX-dependent inflammatory signature is remarkably conserved in human cutaneous melanoma biopsies, arguing for COX activity as a driver of immune suppression across species. Pre-clinical data demonstrate that inhibition of COX synergizes with anti-PD-1 blockade in inducing eradication of tumors, implying that COX inhibitors could be useful adjuvants for immune-based therapies in cancer patients.